

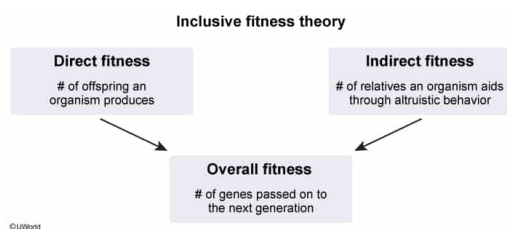
Female *S. beldingi* ground squirrels tend to remain near their birthplace and kin for life. *S. beldingi* females are also more likely than males to give alarm calls when they spot a predator, even though this behavior draws the predator's attention to the individual who produced the alarm. Which concept is LEAST relevant to this phenomenon?

- ✓ ☒ A. Autocommunication [54%]
☐ B. Altruism [13%]
☐ C. Indirect fitness [12%]
☐ D. Evolutionary game theory [18%]

Correct

54% answered correctly

Explanation:



Autocommunication occurs when a message sender is also the receiver. For example, dolphins echolocate by perceiving how the click sounds they have emitted echo back to them. Predator warning calls, communication that alerts *other group members* (not the individual making the call), is not relevant to autocommunication.

(Choice B) Altruism is behavior that **benefits others** in a group at a potential **cost or risk to oneself**. For example, sounding an alarm call in response to a predator sighting benefits the group of *S. beldingi* squirrels, but places the individual who made the call at risk by drawing the predator's attention.

(Choice C) An individual's **inclusive fitness** is the sum of its **direct fitness** (own reproduction) and **indirect fitness** (cooperative behavior that aids kin). Thus, inclusive fitness serves as a metric for an individual's total evolutionary success. Female *S. beldingi* squirrels, who remain near their birthplace and live by many relatives, engage in altruistic behaviors that maximize the chance that kin will survive (ie, indirect fitness).

(Choice D) Game theory mathematically evaluates the **relative success** of particular **strategies** (behavioral phenotypes) over time. When the evolutionary benefit of a social behavior outweighs the cost of that behavior (eg, a warning call puts an individual at risk but increases the likelihood that genes will be passed on), game theory predicts that the behavioral phenotype will persist.

Educational objective:

Altruism is behavior that is carried out for the benefit of others at a cost or risk to the individual.

Altruism can be explained by the theory of inclusive fitness, which states that an individual's overall genetic success is partially derived from the success of genetic relatives, and by game theory, which evaluates the relative successes of particular behavioral phenotypes, such as altruistic alarm calls, over time.